The next positional norm and class maps at fifty-three

E.119. Complete lower class and unit maps, and the exact next coinvariants at 53 [Proved]

Fix $\zeta^2+\zeta+1=0$, $\xi^3=\zeta$, $\alpha^3=53$, and compatible β_d with

$$\begin{aligned} F=\mathbb{Q}(\xi),\quad M=\mathbb{Q}(\zeta,\alpha),\quad \beta_1=\alpha\xi,\quad \beta_{d+1}^3=\beta_d,\\ L_d=\mathbb{Q}(\beta_d),\quad B_d=L_d(\xi),\quad S=\{\text{primes over } 3,53\},\quad A(E)=\mathrm{Cl}_S(E)\{3\}. \end{aligned} \tag{E119.1}$$

Thus $B_1=F(\alpha)$, $[B_1:\mathbb{Q}]=18$, $[L_1:\mathbb{Q}]=6$, and $\beta_d^{3^d}=53\zeta$. Write $\widehat{U}_E=\varprojlim_n \mathcal{O}_{E,S}^{\times}/(\mathcal{O}_{E,S}^{\times})^{3^n}$. The actual lower class groups are

$$\begin{aligned} A(B_1)&=C_3^2, & \mathrm{Cl}(B_1)\{3\}&=C_9\oplus C_3^2,\\ A(L_1)&=C_3^2, & \mathrm{Cl}(L_1)\{3\}&=C_9\oplus C_3. \end{aligned} \tag{E119.2}$$

Identify $A(B_1)$ with $I=\{(e_0,e_1,e_2)\in\mathbb{F}_3^3 : e_0+e_1+e_2=0\}$, using the first two coordinates, and use the certified L_1 ideal coordinates specified in the proof. The horizontal ideal maps are

$$\begin{aligned} N_{B_1/L_1}(e_0,e_1)&=(0,e_0+2e_1), & j_{B_1/L_1}(a,b)&=(a,a),\\ \mathrm{Art}_{B_1/L_1}(a,b)&=2a, & \ker j_{B_1/L_1}&=\{(0,b):b\in\mathbb{F}_3\}. \end{aligned} \tag{E119.3}$$

The complete horizontal unit maps have the explicit matrices and inverse representatives (E119.15)–(E119.20). In particular

$$\begin{aligned} [\widehat{U}_M : N_{B_1/M}\widehat{U}_{B_1}]&=9, & \ker N_{B_1/M}&\simeq C_3\oplus\mathbb{Z}_3^8,\\ N_{B_1/L_1}\widehat{U}_{B_1}&=\widehat{U}_{L_1}, & \ker N_{B_1/L_1}&\simeq C_3\oplus\mathbb{Z}_3^{10}. \end{aligned} \tag{E119.4}$$

Let σ be the actual vertical automorphism $\beta_2\mapsto\zeta\beta_2$, fixing ξ on B_2. The complete next coinvariants, their norms and the ideal-extension maps followed by the quotient projection are

$$\begin{aligned} Q_B:=A(B_2)/(\sigma-1)A(B_2)&=C_9\oplus C_3^3,\\ N_B(x_0,x_1,x_2,x_3)&=(2x_3,x_0+2x_3), & \bar{j}_B(a,b)&=(6a+3b,0,0,0),\\ Q_L:=A(L_2)/(\sigma-1)A(L_2)&=C_9\oplus C_3,\\ N_L(x,y)&=(-x,-x+y), & \bar{j}_L(a,b)&=(6a,0). \end{aligned} \tag{E119.5}$$

The targets of N_B, N_L are the respective C_3^2 in (E119.2), so these formulas reduce their outputs modulo three. Both norms are onto, with kernels C_3^3 and C_3, respectively. Every coordinate map in (E119.5) is obtained from actual ideal Artin maps by the retained integral Smith transformations and their inverses. In particular, the actual capitulation kernels satisfy

$$\begin{aligned} K_B:=\ker\{A(B_1)\xrightarrow{j}A(B_2)\}&\subseteq\{(a,a):a\in\mathbb{F}_3\},\\ K_L:=\ker\{A(L_1)\xrightarrow{j}A(L_2)\}&\subseteq\{(0,b):b\in\mathbb{F}_3\},\\ [\widehat{U}_{B_1}:N\widehat{U}_{B_2}]=3^5|K_B|,&\qquad [\widehat{U}_{L_1}:N\widehat{U}_{L_2}]=3|K_L|. \end{aligned} \tag{E119.6}$$

The field-norm subgroups inside these two lower unit completions have indices 9 and 1. Their quotients by actual completed S-unit norms therefore have orders $3^3|K_B|$ and $3|K_L|$.

The exact upper groups still occur in the sequences

$$\begin{aligned} 0\longrightarrow(\sigma-1)A(B_2)\longrightarrow A(B_2)\longrightarrow Q_B\longrightarrow 0,\\ 0\longrightarrow(\sigma-1)A(L_2)\longrightarrow A(L_2)\longrightarrow Q_L\longrightarrow 0. \end{aligned} \tag{E119.7}$$

No value of either left-hand term is assigned by this statement. The maps (E119.1) identify $L_2 \to L_1$ with the literal root-cubing transition of the $j = 1$ positional factor in E.46. Thus (E119.5)–(E119.7) give an unconditional finite transition for the remaining 53 norm system, together with its precise remaining kernel. The manuscript anchor is §51.16, fols. 733v–736v, scans 172–178, through the actual tower comparisons of E.46 and E.114.

Proof. All ideal class groups below are taken in their three-primary parts when a finite three-group is displayed. Completion of a finitely generated multiplicative group means tensoring its free part with $\mathbb{Z}_3$ and retaining its three-primary roots of unity. These groups are compact. The completion of a whole field's multiplicative group has the different, explicitly prodiscrete topology used in E.46; the comparison there is induced by the literal inclusion of S-units.

The fields and their labeled local maps. The exact equations show that L_1 has polynomial $X^6 + 53X^3 + 2809$ and that B_1 is the compositum of $X^3 - 53$ and $\Phi_9(X)$. The actual primitive element $x = \xi - \alpha$ has polynomial

$$x^{18} + 321x^{15} + 41505x^{12} + 2826974x^9 + 112000083x^6 + 2393775690x^3 + 23467349647. \quad \text{(E119.8)}$$

All tensor-coordinate calculations use $(\alpha^r \xi^s)_{0 \le s \le 5,\, 0 \le r \le 2}$ in this order. The executed polynomial and basis identities verify each embedding.

Since $53 = -1 + 54$, the logarithm on $1 + 3\mathbb{Z}_3$ gives $53 \in (\mathbb{Q}_3^\times)^9$. Let a_0 be its cube root congruent to -1 modulo nine; $v_3(a_0 + 1) = 2$ and a_0 is itself a cube. The three primes q_j of B_1 over three are labeled by $\alpha \mapsto a_0 \zeta^j$, $j = 0, 1, 2$, with ξ fixed. Their completions are $\mathbb{Q}_3(\xi)$. There are three primes t_i over 53, all with residue field $\mathbb{F}_{53^2}$. The extension B_1/F is split at three and totally ramified at these three 53-primes; B_1/M is totally ramified at its three 3-primes and split at 53. The extension B_1/L_1 is unramified and split at all S-primes. Indeed its local map at three is already the equality $L_{1,3} = \mathbb{Q}_3(\xi)$, and at 53 the residue field contains μ_9 because $9 \mid 53^2 - 1$.

The vertical extension B_2/B_1 is totally ramified at all six S-primes. At 53 this follows from the valuation one of β_1. At q_j, its Kummer equation is $Z^3 = a_0 \xi^{1+3j}$ and its local field is $\mathbb{Q}_3(\zeta_{27})$ over $\mathbb{Q}_3(\xi)$. Here $\beta_1 + 1$ is an actual norm uniformizer: it is the norm of $\beta_2 + 1$, and its local valuation is one. The extension L_2/L_1 likewise has exactly two totally ramified S-places. The displayed equations have discriminants supported at $3, 53$, which excludes any further ramification.

The entire first horizontal S-class group. The certified field F has class number one. Its seven complete S-unit generators are

$$\xi,\ 1 + \xi,\ 1 + \xi^2,\ -1 + \xi^4,\ -4 + \xi^4 + \xi^5,\ 3 + 2\xi + 3\xi^2 + 3\xi^3 - \xi^4 + \xi^5,\ -4 - \xi + \xi^2 - \xi^4. \quad \text{(E119.9)}$$

Every generator has zero cubic local character for B_1/F at all three 53-primes; the retained finite-field reductions check the three rows separately. At three the extension is split. The idele class quotient with local norm groups at S and local units outside S is therefore C_3^3. Its restriction to $\mathrm{Gal}(B_1/F)$ is the sum of its three coordinates. To identify its kernel with class coinvariants, let $H_{B_1}^S$ be the S-Hilbert three-class field. A decomposition group above a totally ramified 53-prime supplies a section of $\mathrm{Gal}(H_{B_1}^S/F) \to C_3$, since that prime splits completely in $H_{B_1}^S/B_1$. Abelianizing this split extension identifies the kernel with $A(B_1)/(g-1)A(B_1)$, where $g(\alpha) = \zeta\alpha$ and $g(\xi) = \xi$. Thus this coinvariant group is exactly I.

For completeness, the idele construction used here is the actual global reciprocity map whose restriction to a local multiplicative group is local reciprocity [1, V,Proposition5.2 and

Theorems5.3 and5.5]. For an abelian extension of F, pullback to B_1 is unramified outside S precisely when its local units outside S are killed; it splits at each place in S precisely when the local norm subgroup is killed. This proves both inclusions between that idele quotient's class field and the maximal subfield of $H^S_{B_1}$ abelian over F.

Now let c be complex conjugation fixing α. Its fixed field is $R = \mathbb{Q}(\alpha, \xi + \xi^{-1})$, of signature $(3, 3)$. The literal compositum gives the polynomial

$$X^9 - 9X^7 + 162X^6 + 27X^5 + 459X^4 + 7290X^3 + 1458X^2 + 51030X + 156033. \quad \text{(E119.10)}$$

The successful `bnfcertify(R,1)` gives a natural class-presentation upper quotient C_3. Its five actual S-prime classes in this quotient are $(1), (1), (0), (0), (0)$. The certificate also retains and checks the full principal ideal relation for each of these five coordinates and the third power of the sole class generator. Consequently the true $A(R)$ is zero. This conclusion uses exactly the flag-one class-generation assertion; unit completeness is proved separately below. The official flag-one contract is [3, `bnfcertify`, flag1]. The actual quadratic maps satisfy $Nj = 2$ and $jN = 1 + c$. They give $1 + c = 0$ on $A(B_1)$, so $c = -1$. The field identity $cgc = g^{-1}$ now gives $g = g^{-1}$ on this three-group. Together with $g^3 = 1$ it gives $g = 1$. This proves $A(B_1) = I$, with the same ideal map used above.

A literal S-Hilbert field and the horizontal class maps. Set $h(\xi) = \xi^4$ and

$$\begin{aligned} \pi_0 &= -1 + 3\xi - 3\xi^2 + 3\xi^4, \\ \pi_1 &= -1 - 3\xi + 3\xi^2 + 3\xi^5, \qquad \pi_2 = -1 - 3\xi^4 - 3\xi^5, \\ h(\pi_j) &= \pi_{j+1}, \qquad \pi_0\pi_1\pi_2 = 53. \end{aligned} \quad \text{(E119.11)}$$

Indices on π_j are modulo three. Their valuation matrix at the three 53-primes is $\left(\begin{smallmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{smallmatrix}\right)$. At three put $\lambda = 1 - \xi$ and $w = -1 + 2\lambda^2 + \lambda^3$. The exact polynomial calculation gives $v_\lambda(w^3 - \pi_0) = 13 > 12 = 2v_\lambda(3w^2)$; Hensel's lemma gives a local cube root. Conjugating by h proves this for all three radicals. At every 53-prime the residue of $\pi_j/53^{v(\pi_j)}$ has cubic character zero. Thus

$$\mathcal{H} = F(\gamma_0, \gamma_1, \gamma_2), \quad \gamma_j^3 = \pi_j, \quad \gamma_0\gamma_1\gamma_2 = \alpha, \qquad \mathcal{H} = H^S_{B_1}. \quad \text{(E119.12)}$$

The independent prime valuations prove $[\mathcal{H} : F] = 27$. The local calculations prove that $\mathcal{H}/B_1$ is unramified and S-split. Its degree nine equals $|A(B_1)|$, proving the last equality by the class-field correspondence [1, V,Theorem3.6 and Example3.9]. All π_j are fixed by c; choose their real cube roots. Then c negates all Kummer phases, while h cyclically permutes them.

The first two Kummer phases give the coordinates of I used in the statement. The actual prime data

p	ξ	α	β_1	(e_0, e_1, e_2)	$[\mathfrak{p}]_{A(L_1)}$
163	53	6	155	$(0, 2, 1)$	$(0, 1)$
163	40	6	77	$(0, 1, 2)$	$(0, 2)$
163	38	6	65	$(2, 0, 1)$	$(0, 2)$

(E119.13)

give the norm row in (E119.3); every entry is a literal residue in $\mathbb{F}_{163}$. The ideal norm sends each prime to its contraction to the displayed residue-degree power, as in the proved Frobenius square [1, V,Proposition3.3]. The certified ordinary class coordinates of L_1 are $C_9 \oplus C_3$, with its 3-prime $(6, 0)$ and its 53-prime zero; reducing the first coordinate modulo three gives $A(L_1)$. Two actual primes over seven have ordinary coordinates $(1, 0)$ and $(7, 1)$. Both have residue degree one and act on ξ by $\xi \mapsto \xi^7$, giving the Artin row $2a$ in (E119.3).

The whole group $\mathrm{Gal}(\mathcal{H}/L_1)$ is

$$\{(e,b)\in\mathbb{F}_3^3\rtimes\mathbb{F}_3 : e_0+e_1+e_2+b=0\},\qquad (e,b)(f,d)=(e+h^b f,b+d). \tag{E119.14}$$

Here h^b acts by cyclic permutation and $\tau=(-e_0,1)$ fixes $\beta_1=\alpha\xi$. Its cube is the phase $-(1,1,1)$. The transfer from this group to I is zero on I, since $1+h+h^2$ kills I, and sends τ to τ^3. The Artin coordinate of τ is one, hence its first L_1 coordinate is two. This gives exactly $j(a,b)=(a,a)$, including its sign. The inclusion/transfer square follows from the proved restriction/corestriction compatibility of the reciprocity cup product [1, III,Theorem3.1 and its proof]; the calculation here specifies the actual transfer on every element. More explicitly, on $\widehat{H}^{-2}(G,\mathbb{Z})=G^{\mathrm{ab}}$, restriction to a subgroup is group transfer and corestriction is the map induced by subgroup inclusion. On $\widehat{H}^0(G,E^\times)$, the corresponding maps are field inclusion and field norm. Cupping with the fundamental class therefore gives inclusion–transfer and norm–restriction for the compatible local Artin maps. The product of these local maps in V,Proposition5.2 gives the stated global ideal maps.

Complete units, divisors and inverse norm coordinates. Let $U_1,\dots,U_{15}$ be the literal elements in the `tensor_unit_columns` table of `e53_horizontal_norm_matrix.out`. This table consists of eighteen rational coefficients per element in the tensor basis following (E119.8); it is part of the retained certificate. In particular $U_1=\xi$, $U_2=-1+\xi-\xi^3$, $U_3=-1-\xi^3+\xi^4$, and $U_{12}=-1+\xi^4$. The proof uses every full principal ideal, checked against its displayed S-divisor. It does not use the candidate upper BNF's class number or unit completeness.

In the analytic order $(q_0,q_1,q_2,t_1,t_2,t_3)$, the only nonzero S-valuation columns are $v(U_{10})=(0,0,3,0,0,0)$, $v(U_{11})=(0,3,0,0,0,0)$, $v(U_{12})=(1,1,1,0,0,0)$ and $v(U_{12+i})=e_{3+i}$, $1\le i\le 3$. The analytic order of the 3-primes is $(3,2,1)$ in the retained PARI prime list. Use the complete certified M basis from E.116, $(\zeta,f_1,f_2,q_1,q_2,q_3,\alpha)$, where the q_i are the literal principal 3-prime generators in that PARI order. The complete norm matrix is

$$N_M=\begin{pmatrix}
1&2&2&2&0&1&2&2&2&1&1&1&1&1&1\\
0&0&0&0&0&0&0&1&0&2&2&2&0&0&0\\
0&0&0&0&0&0&0&1&1&1&1&1&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&1&0&0&0\\
0&0&0&0&0&0&0&0&0&3&0&1&0&0&0\\
0&0&0&0&0&0&0&0&0&0&3&1&0&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&1&1&1
\end{pmatrix}. \tag{E119.15}$$

The first row is modulo three; the other rows lie in $\mathbb{Z}_3$. Multiplication by -1 disappears under this completion by its literal compatible odd roots. Before completion, the retained PARI objects contain the full signed algebraic norms.

Here is an independent proof that the matrix gives the entire norm image and that the listed elements are complete. For a cyclic extension E/D unramified outside S, the map

$$H^1(\langle\sigma\rangle,\mathcal{O}_{E,S}^\times)\longrightarrow\ker\{\mathrm{Cl}_S(D)\to\mathrm{Cl}_S(E)\},\quad [\sigma x/x]\longmapsto[D_0],\quad \mathrm{div}_{\notin S}(x)=jD_0 \tag{E119.16}$$

is an isomorphism. Indeed Hilbert 90 supplies x; its divisor is invariant, and outside S every invariant divisor is an extended base divisor, because there is no ramification. Changing x by a base element changes D_0 by a principal divisor; changing its cocycle by a unit coboundary does the same. Conversely a principal extended divisor supplies x and its unit cocycle. These constructions are mutual inverses. This is the explicit outside-S version of the ideal/Hilbert90 maps in [2, Proposition1 and full ideal proof].

E.116 gives $A(M)=0$. For B_1/M, (E119.16) therefore has zero three-primary part. The S-unit Herbrand quotient is $3^3/3=9$, since exactly the three 3-places have local degree

three; see the complete lattice proof in [1, VII,Proposition3.1]. Thus the true completed unit-norm index is nine. Adjoining the torsion relation $3e_1$ to (E119.15) gives an image HNF of determinant nine. Every column is an actual norm, so the candidate image is the entire image. At each 3-place the relative residue degree is one, hence the norm preserves its normalized valuation. It follows that the full principal S-divisor lattice of B_1 is exactly

$$\{(a_0,a_1,a_2,b_1,b_2,b_3)\in\mathbb{Z}_3^6 : a_0\equiv a_1\equiv a_2 \pmod 3\}. \tag{E119.17}$$

The valuation columns already listed give its converse inclusion.

The first nine elements are ordinary units, including ξ. Their nine residue-character rows at the actual prime labels

$$(7,1,3),(7,2,3),(7,3,3),(7,4,3),(7,5,3),$$
$$(11,1,6),(163,1,1),(163,2,1),(397,1,1)$$

have determinant -17. The labels mean rational prime, actual prime-list index, and residue degree. Both the matrix and its full inverse modulo three are retained. The ordinary unit rank is eight; the three-primary roots of unity are exactly μ_9. Indeed $\mu_{27}\subset B_1$ would force the degree-eighteen field B_1 to equal $\mathbb{Q}(\zeta_{27})$, contradicting its ramification at 53. Thus the determinant proves that these nine classes form a basis modulo cubes of the full ordinary unit group. Nakayama's lemma on its finite-rank completion proves completeness. With (E119.17), all fifteen columns give $\widehat{U}_{B_1}=C_9\oplus\mathbb{Z}_3^{14}$.

For a target $\zeta^z f_1^x f_2^y q_1^{a_1}q_2^{a_2}q_3^{a_3}\alpha^w$ with $a_1\equiv a_2\equiv a_3 \pmod 3$, put $r=a_1$, $s=(a_2-a_1)/3$, $t=(a_3-a_1)/3$, $d=r+s+t$, $x'=x-2d$, $y'=y-d$. An actual inverse representative and the full kernel are

$$\begin{aligned} &U_{12}^r U_{10}^s U_{11}^t U_8^{x'} U_9^{y'-x'} U_{13}^w \xi^{[z-2y'-d-w]_3},\\ \ker N_M=\langle\xi^3\rangle\ \oplus\ &\langle U_2\xi^{-2},U_3\xi^{-2},U_4\xi^{-2},U_5,U_6\xi^{-1},U_7\xi^{-2},U_{14}/U_{13},U_{15}/U_{13}\rangle_{\mathbb{Z}_3}. \end{aligned} \tag{E119.18}$$

Here $[z]_3$ means the representative $0,1,2$ of the torsion coordinate, and every other exponent is three-adic. Reading the eight free nonpivot coordinates and then the exponent of ξ^3 gives the inverse kernel coordinates. This is a continuous coset representative; the map $C_3\to C_9$ taking these digits has its actual carry in $\langle\xi^3\rangle$.

The complete L_1 unit basis is $(\zeta,u_1,u_2,q,\beta_1)$, where

$$\begin{aligned} u_1&=-7+\beta_1^2-(32/53)\beta_1^3+(7/53)\beta_1^4-(1/53)\beta_1^5,\\ u_2&=-7+7\beta_1+\beta_1^2-(32/53)\beta_1^3+(2/53)\beta_1^5,\\ q&=3600+1116\beta_1+199\beta_1^2+(3777/53)\beta_1^3+(411/53)\beta_1^4-(91/53)\beta_1^5. \end{aligned} \tag{E119.19}$$

Its two S-valuation columns are $(3,0)$ and $(0,1)$. In this basis the actual horizontal norm matrix is

$$N_L=\begin{pmatrix} 1&2&2&1&2&0&1&0&0&2&1&2&0&1&1\\ 0&0&0&1&2&2&1&0&-1&2&1&2&0&1&-1\\ 0&0&0&2&1&1&2&-1&-1&2&3&2&-1&1&0\\ 0&0&0&0&0&0&0&0&0&1&1&1&0&0&0\\ 0&0&0&0&0&0&0&0&0&0&0&0&1&1&1 \end{pmatrix}. \tag{E119.20}$$

For a target (z,x,y,c,d), set $x'=x-2c$, $y'=y-2c+d$. The representative $U_{10}^cU_{13}^dU_9^{-x'}U_8^{x'-y'}\xi^{[z-2c]_3}$ maps exactly to that target. Subtracting this representative from the ten nonpivot source generators $U_2,\dots,U_7,U_{11},U_{12},U_{14},U_{15}$ gives a free kernel basis; ξ^3 gives its C_3 summand. The forward and inverse integer matrices are retained. They prove both topological isomorphisms in (E119.4).

The divisor lattice (E119.17) has index nine, so the subgroup of $\mathrm{Cl}(B_1)\{3\}$ generated by S has order nine. Together with $A(B_1) = C_3^2$ this proves that the ordinary three-class group has order 81. Three literal ideals G_1, G_2, G_3, moved outside S by retained exact principal corrections, satisfy the verified relations $G_1^9 = (f_1)$, $G_2^3 = (f_2)$, $G_3^3 = (f_3)$. Their full HNF ideals, principal factors and factor-prime proofs are retained. Their Kummer genus columns and the actual S-prime coordinates are

$$\begin{pmatrix} 1 & 0 & 2 \\ 0 & 0 & 2 \\ 2 & 0 & 2 \end{pmatrix}, \qquad \begin{pmatrix} 3 & 0 & 6 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}. \tag{E119.21}$$

The S columns here are in the original prime-list order. The first matrix spans $A(B_1)$, and the second shows that the generated subgroup contains the entire S-prime subgroup. It has order at least 81; the three principal power relations bound its order by $9 \cdot 3 \cdot 3 = 81$. This proves its generation and exact type in (E119.2). The attempted upper BNF certification terminated by timeout; its source and actual exit output are retained. None of the order, generation or saturation deductions above uses its success.

Six actual local characters at the next level. Let $\chi_v(u)$ be the exponent on a cube root of β_1 under arithmetic local reciprocity of u. At q_j, write $v = v_{q_j}(u)$. The actual cyclotomic norm square gives

$$\begin{aligned} \chi_{q_j}(u) &= -\frac{N_{\mathbb{Q}_3(\xi)/\mathbb{Q}_3}(u/(\beta_1+1)^v) - 1}{9} \pmod 3, \\ \chi_{t_i}(u) &= -\log_\zeta \left(\overline{u/\beta_1^{v_{t_i}(u)}}^{(53^2-1)/3} \right). \end{aligned} \tag{E119.22}$$

For the first formula, $\beta_1 = a_0 \xi^{1+3j}$ locally; a cube root of a_0 is already in $\mathbb{Q}_3$. Arithmetic reciprocity on cyclotomic units raises roots of unity to the inverse absolute norm. On the root of ξ^{1+3j}, the factor $1+3j$ is one modulo three. This gives the first sign and the denominator nine. The chosen norm uniformizer has character zero. The second formula is the tame symbol with first argument u, second argument β_1; the local uniformizer is again a norm. Both formulas use the same global ζ and the same local reciprocity normalization. The norm/inclusion square used in these statements follows from the proved cup-product compatibility in [1, III,Theorem3.1].

On the complete fifteen columns the first three character rows are all c_q and the last three are all c_t, where

$$\begin{aligned} c_q &= (0,0,0,0,0,0,0,0,0,1,1,0,1,1,1), \\ c_t &= (0,0,0,0,0,0,0,2,1,0,0,0,0,0,0). \end{aligned}$$

Their rank is exactly two and their column sums are zero. All entries are retained with their local valuation corrections and the checked three-adic precision. Since units at unramified places are local norms, the cyclic Hasse norm theorem, including its injectivity proof [1, VIII,Theorem3.1], identifies the common kernel with the field-norm subgroup inside the completed S-units. Its index is $3^2 = 9$. This also proves the completion assertion: the quotient is killed by three, so the finite character map has the identical kernel after completion and its image remains unchanged.

The actual next genus group over B_1. Let Γ_B be the Galois group over B_1 of the maximal subfield of $H_{B_2}^S$ abelian over B_1. The same local-to-global Artin maps as above

identify it with the three-primary part of

$$J_{B_1}\Big/B_1^{\times}\left(\prod_{v\in S}N(B_{2,w}^{\times})\right)\left(\prod_{v\notin S}\mathcal{O}_{B_{1,v}}^{\times}\right). \tag{E119.23}$$

Every S-place is totally ramified, so the local factors have order three. The exact sequence from S-units to these six factors and then to $A(B_1)$ gives $|\Gamma_B| = 3^{6-2}\cdot 9 = 729$. This is an actual idelic quotient, with ideals mapped by their uniformizer ideles. It therefore retains all class-extension information needed to distinguish exponent nine from exponent three.

Choose the six local generators so that each restricts to the fixed generator of $\mathrm{Gal}(B_2/B_1)$, followed by the three outside-S ideal generators G_i in (E119.21). For $G_i^{n_i} = (f_i)$ the relation is $n_iG_i + \sum_v \chi_v(f_i)E_v = 0$. For $Q_v = (f_v)\prod_i G_i^{c_{iv}}$ it is $\sum_i c_{iv}G_i - \sum_w \chi_w(f_v)E_w = 0$. These signs follow by applying the global product formula to the principal ideles f_i, f_v; the former have divisor n_iG_i outside S, whereas the latter have divisor $-\sum_i c_{iv}G_i$ outside S.

Thus an integral presentation matrix is

$$R_B = \begin{pmatrix} 3I_6 & C_{\mathrm{ord}} & R_c & R_s \\ 0 & 0 & \mathrm{diag}(9,3,3) & C_s \end{pmatrix}, \quad R_c = \begin{pmatrix} 0&1&0\\0&1&0\\0&1&0\\0&0&1\\0&0&1\\0&0&1 \end{pmatrix}, \tag{E119.24}$$

where C_{ord} is the first nine character columns, C_s is the second matrix in (E119.21), and

$$R_s = \begin{pmatrix} -2&-2&-1&-1&-1&-1\\0&0&-2&-1&-1&-1\\-2&-2&-1&-1&-1&-1\\-1&0&-1&0&0&0\\-2&0&0&0&0&0\\0&0&-2&0&0&0 \end{pmatrix}. \tag{E119.25}$$

All principal factors defining these columns are retained as products of literal algebraic numbers; equality of their ideals is checked by exact prime-valuation cancellation. Every rational prime used in those cancellations is proved prime by the program. Factored generators avoid an expansion that overflowed the stack in a preserved earlier run; the exact factorized equality supplies the same principal divisor map.

The HNF of (E119.24) is

$$H_B = \begin{pmatrix} 3&0&1&0&0&0&0&0&0\\0&3&1&0&0&0&0&2&0\\0&0&1&0&0&0&0&0&0\\0&0&0&3&0&1&2&0&0\\0&0&0&0&3&1&1&0&0\\0&0&0&0&0&1&0&0&0\\0&0&0&0&0&0&3&0&0\\0&0&0&0&0&0&0&1&0\\0&0&0&0&0&0&0&0&3 \end{pmatrix}, \qquad \rho_B = (1,1,1,1,1,1,2,1,0). \tag{E119.26}$$

The row ρ_B is the actual restriction to B_2. Its last three entries are the cubic residue symbols of β_1 on the actual factors of G_1, G_2, G_3. It annihilates every relation modulo three. The retained unimodular Smith maps give $\Gamma_B = C_9 \oplus C_3^4$.

For an entirely specified inverse to the kernel coordinates, put $K_B^{\mathrm{lat}} e_1 = 3e_1$ and $K_B^{\mathrm{lat}} e_j = e_j - \rho_{B,j} e_1$ for $2 \le j \le 9$. Its image is the full inverse image in $\mathbb{Z}_3^9$ of $\ker \rho_B$. With $H'_B = (K_B^{\mathrm{lat}})^{-1} H_B$, the retained unimodular matrices P_B, Q'_B satisfy

$$P_B H'_B Q'_B = \mathrm{diag}(9,3,3,3,1,1,1,1,1), \quad z \longmapsto K_B^{\mathrm{lat}} P_B^{-1} z, \quad x \longmapsto P_B (K_B^{\mathrm{lat}})^{-1} x. \quad \text{(E119.27)}$$

The displayed maps are mutual inverses modulo the actual relation lattices. The executable certificate checks both compositions for all 243 kernel classes and all 729 classes of Γ_B.

To identify this kernel with the exact group in (E119.5), use a decomposition group of $H_{B_2}^S/B_1$ above any totally ramified S-prime. Its restriction to B_2 is an isomorphism onto C_3, because the next extension is S-split. Hence $\mathrm{Gal}(H_{B_2}^S/B_1) = A(B_2) \rtimes C_3$. Its abelianization has kernel $A(B_2)/(\sigma - 1)A(B_2)$. This gives exactly the first quotient map of (E119.7). Norm to $A(B_1)$ kills the six local generators and sends G_i to the first two rows of (E119.21). Composing that matrix with (E119.27) gives N_B in (E119.5). Extension to coinvariants sends an ideal generator to three times its lifted idele class: in the split semidirect product this is the transfer followed by abelianization. Different lifts differ by local elements of exponent three, so this map is independent of the lift. Lifting $(a,b) \in A(B_1)$ by $G_1^{a-b} G_3^{2b}$ and applying (E119.27) gives $\bar{j}_B(a,b) = (6a + 3b, 0, 0, 0)$. These maps satisfy $N_B \bar{j}_B = 3$ and $\bar{j}_B N_B = 3$, in their respective groups.

The direct positional transition. For L_2/L_1, the two characters on every one of the five complete S-unit generators (E119.19) are zero. The wild formula is (E119.22) with the exact global norm $N_{L_1/\mathbb{Q}}(u)$ divided by $2757^{v_3(u)}$; here $2757 = 53^2 - 53 + 1$ is the norm of $\beta_1 + 1$. The tame formula is unchanged. The certified two ordinary ideal generators of orders nine and three have zero class-power characters. Their outside-S Frobenius characters are both one. The S-prime relations have lower class columns $(6,0),(0,0)$ and negative local character columns $(-1,-2),(0,0)$. The resulting presentation, restriction and norm-kernel lattice are

$$R_L = \begin{pmatrix} 3 & 0 & 0 & 0 & -1 & 0 \\ 0 & 3 & 0 & 0 & -2 & 0 \\ 0 & 0 & 9 & 0 & 6 & 0 \\ 0 & 0 & 0 & 3 & 0 & 0 \end{pmatrix}, \quad \rho_L = (1,1,1,1),$$
$$H_L = \begin{pmatrix} 3 & 0 & 1 & 0 \\ 0 & 3 & 2 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}, \quad K_L^{\mathrm{lat}} = \begin{pmatrix} 3 & -1 & -1 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad \text{(E119.28)}$$

The same actual idelic and decomposition-group maps identify $\Gamma_L = C_9 \oplus C_3^2$ and its kernel $Q_L = C_9 \oplus C_3$. The Smith transformations of $(K_L^{\mathrm{lat}})^{-1} H_L$ have diagonal $(9,3,1,1)$. Composing their inverse with the projection to the two ideal coordinates gives $N_L(x,y) = (-x, -x + y)$; composing their forward map with three times the two lifted ideal generators gives $\bar{j}_L(a,b) = (6a, 0)$. All 27 kernel classes and all 81 full classes, with both inverse maps, are checked. The explicit norm kernels are $\{(3r, x_1, x_2, 0)\} \simeq C_3^3$ for B and $\{(3r,0)\} \simeq C_3$ for L.

The retained unit defect and tower comparison. The actual class norm is onto in both vertical extensions. One proof with its precise map is to intersect the lower S-Hilbert three-class field with the cyclic upper field. That intersection is trivial over the base: the latter is ramified at S, while the former is split there. For any lower class, Chebotarev applied to their compositum gives a prime with that class and trivial upper Frobenius. Its degree-one upper prime has that exact ideal norm. Equivalently, the norms in (E119.5) already give all

nine lower classes. The full ideal-extension map followed by the quotient in (E119.7) is the displayed $\overline{j}$; therefore its actual kernel injects into the two lines in (E119.6).

Apply the explicit isomorphism (E119.16) to these two vertical extensions. Their S-unit Herbrand quotients are $3^6/3 = 3^5$ and $3^2/3 = 3$, respectively. This proves the two exact index formulas in (E119.6). Tensoring the finite-rank unit sequences with $\mathbb{Z}_3$ is exact, and finite cyclic group cohomology commutes with this flat tensor product; hence these are the stated completed indices as well. The field-norm indices follow from the complete local character matrices and the cyclic Hasse norm theorem. Dividing the two indices proves the asserted orders of the remaining unit defects. This retains the actual capitulation map instead of replacing a unit norm by a field norm.

Finally the evaluation-at-a-root Kummer–Shapiro map of E.46 identifies the root-cubing pushforward with N_{L_2/L_1}, using the literal embeddings $\beta_1 = \beta_2^3$ in (E119.1). Its extension to the auxiliary field is induced by $L_d \hookrightarrow B_d$; horizontal norms commute with the vertical norms by the tower identity for the product of conjugates. On divisors these are the maps $\mathfrak{P} \mapsto \mathfrak{p}^{f(\mathfrak{P}/\mathfrak{p})}$; on classes and units they are the maps computed above. The obstruction to replacing the exact upper class group by its evaluated coordinates is precisely the left term of (E119.7). Evaluating that term and the resulting actual capitulation is the remaining step before these finite transition formulas can be iterated as a full positional norm-system classification. □

References

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